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ANRF AISEHack | 2026 Finale Presentation Theme: Pollution Forecasting

Team LuVW

Gautam Gandhi, Dhimant Bhuvra,

Manas Inamdar, Jainil Bavishi

Best Score: 0.8877 | Model: ~5.6M params
Trained on Kaggle T4x2 GPU in ~1.5 hours

Key Contributions

f1

EnhancedUNet: 4-level Residual U-Net + SE Attention

Multi-scale encoder-decoder with coordinate channels.
4-component episode-aware loss function.

f2

2-Phase Curriculum + SWA Ensemble

Phase 1: balanced loss (30 epochs)
Phase 2: episode-heavy (15 epochs)
SWA weight averaging of both phases.

f3

Score: Baseline 0.55 → Best 0.8877

3 iterative submissions. Biggest gain from
episode-aware loss + chronological split.

Modelling - *Strategies*

01

Input Representation

10 timesteps x 16 features $\rightarrow$ 160 channels + 2 coordinate grids.
All features z-score normalised from training data only.

Emission features (PM25, NH3, SO2, NOx, NMVOCs, bio)
provide source-level context for secondary aerosol formation.

01

02

Episode-Aware Loss Design

4-component loss: L1 + GlobalSMAPE + EpisodeSMAPE +
EpisodeCorr.
Directly targets the competition's scoring metric.

02

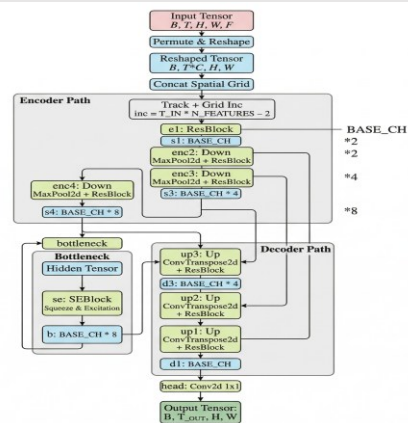
03

SWA Ensemble

Average Phase 1 + Phase 2 weights via SWA.
Recompute BN stats. Ensemble: $0.5 \cdot P2 + 0.5 \cdot SWA$.

03

Architecture: EnhancedUNet



Strengths: Episode-aware loss,
SWA smoothing, SE attention

Weaknesses: Time flattened
as channels, no temporal attention

Key Results – *Why, what and how of the workings?*

S1

What is working?

EnhancedUNet (4-level + SE). 2-phase curriculum training.
Chronological split with 16-step gap. AdamW + CosineAnnealing.
Phase 2 loss weights: L1=0.15, GS=0.20, ES=0.35, EC=0.30

● Implemented

Best score: 0.8877

S2

How is it working? (Quantitative evals)

Baseline: ~0.7780 → Attempt 1 (FNO): 0.7770 → Attempt 2 (UNet): 0.8809 → Best: 0.8877
SSIM: 0.90 | FAC2: 0.95 | Val loss < 0.10 across all 4 seasons

S3

Why is it working? (Reasoning and analysis)

Episode-aware loss directly optimises for competition metric.
Curriculum training handles class imbalance (episodes are rare).
SE block focuses on cpm25, pblh, wind matching atmospheric physics.

S4

Parameters: ~5.6M | Training: ~3h on P100 GPU | Inference: ~30s for 218 samples
Mixed precision (AMP). Batch size 4. Seed 42 throughout.

Ablation Study & Error Analysis

01

Loss Function Impact

Switching from L2 to 4-component episode loss gave the biggest single improvement (+0.10).
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01

02

Data Split Fix

Random split → chronological split removed temporal leakage. Essential for honest validation.

02

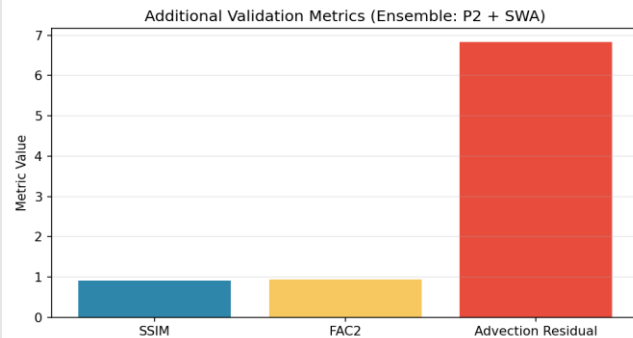
03

What Didn't Work

Wider UNet (base 64), 5th level, temporal attention, multi-seed ensemble, no improvement over SWA.

03

Validation Metrics



Key insight: Loss design matters more than architecture.

SE block: ~+0.01
Curriculum: ~+0.03
SWA: ~+0.01

Details of Relevant Experimentation and Methodology

01

Ablations: FNO→UNet (+0.10), +SE (+0.01), +curriculum (+0.03), +SWA (+0.01)

Tried wider model, 5th encoder level, temporal attention, multi-seed ensemble, none beat SWA approach.

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01

02

Papers: FNO [Li, ICLR 2021], SE-Net [Hu, CVPR 2018], SynCast [arXiv:2510.23977]

FNO: coordinate embeddings. SE-Net: channel attention.
SynCast: domain-aware loss for rare pollution events.

02

03

Claude Code used for: literature review, report drafting, code analysis

Created .md context files from competition pages first.
Used AI for feedback on pipeline, not for writing model code.

03

Links & Thank You

- GitHub: github.com/Dhimant-1001/AISEHack-Phase2-PM25-Forecasting
- Kaggle: kaggle.com/code/dhimantbhuva/notebookc2c8f8d722
- Model: kaggle.com/models/dhimantbhuva/pm25-forecasting-finals submission
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- Thank you!